

NORWEGIAN UNIVERSITY OF SCIENCE AND TECHNOLOGY

IT3010 - 4

RESEARCH-BASED INNOVATION METHODOLOGIES IN COMPUTER AND INFORMATION
SCIENCE

Efficiency of Panopto

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Purpose

Motivation

In recent years, Panopto has become a popular platform for recording and sharing educational lectures, making it a valuable tool for students [7]. However, the usefulness of the platform for students relies heavily on its usability. According to ISO 9241 usability is defined as the "extent to which a system, product or service can be used by specified users to achieve specified goals with effectiveness, efficiency and satisfaction in a specified context of use." [4].

With the rise of remote learning, students are often required to access video lectures on their own time and from various locations. As a result, the usability and efficiency of the platform become even more critical for students, as it directly impacts their ability to access and consume course material [8, 3].

By researching the efficiency of Panopto's usability, one can identify ways to improve the platform's accessibility, responsiveness, and overall user experience, making it easier and more efficient for students to watch lectures and engage with course content.

Furthermore, improved efficiency in usability can have a positive impact on students' academic success. By reducing the time and effort required to access and engage with video lectures, students can focus more on learning and retaining the course material, ultimately leading to better academic performance [2].

In addition to benefiting students directly, research into the efficiency of Panopto's usability can also help educators and instructional designers optimize their course content delivery. By identifying areas where the platform can be improved, educators can ensure that their students have the best possible experience while watching lectures, leading to higher satisfaction rates and better student outcomes.

Research Objective

The objective of this research is in the area of computer science, the structure and effectiveness of an user interface and lastly to add research data to areas we find to be lacking. More specifically to evaluate through research data how to organize and optimize the usage of a video sharing platform for the collective student experience.

Research Question

- How efficient is Panopto as a e-learning tool?

As our chosen research strategy is experiment, read more about this in section Research methods, we propose the following hypothesis to test the research question:

The ineffectiveness of the user interface of Panopto results in students not accepting Panopto as a e-learning tool.

Contributions

Based on the framework provided by March and Smith [5], the purpose of the research is to evaluate a model. Specifically, evaluating the effectiveness, in terms of usability, of Panopto.

	Build	Evaluate	Theorize	Justify
Constructs				
Model		X		
Method				
Instantiation				

Table 1: Showing the contributions of this research paper

To achieve this, the proposed hypothesis will be tested, and the result will be the contribution of this research.

Research methods

Since the purpose of the study is to understand the usability of Panopto in the form of efficiency, holding an experiment is the research strategy that best suits this task. The data-generating method to be used is observations. Since we are focusing on the observation part in a controlled environment we can rule out other strategies like case study and survey. This is due to the nature of trying to understand efficiency. To measure the efficiency, the subjects are given a number of tasks to complete. By measuring how quickly they complete these tasks, we can calculate the time-based efficiency of Panopto [1]. This data that is generated through the experiment is quantitative, that is therefore the data analysis method. The main threat to testing efficiency is how well the subjects know the system beforehand, as this system has been in use for quite some time at NTNU some subjects could have already frequently used it. However, even with that being the case they could still come up with improvement points on how to make the system more efficient.

Participants

As our area of research is based on educational e-learning tools, the participants are to be students at a high education level. Since we as of now only have access to the viewer version of Panopto and not the publisher version, there is a restriction on the types of participants in our experiment. The participants in the experiment would be our group as observers and a group of students as subjects. These students can come from any background as our test would benefit from different technological abilities. These types of subjects are necessary as different technological backgrounds can affect the understanding of technological websites. As this experiment has a narrow scope, the general aspects can still be utilized by others trying to make a video platform for lectures. This research can either be used by Panopto's user interface team, or other video providers wanting to make an efficient platform. Since we are using experiments, it would be helpful to record the computer screen to accurately time the completion of tasks. Therefore, there is little need to think about anonymity and confidentiality as the only thing we are recording is the computer screen, as long as the subjects feel comfortable sharing their technological abilities anonymously. This is not data that can be traced back to people as it would show the difference in technological abilities. To make sure that the screen recordings do not somehow capture personal data from subjects, we will be providing computers to the subjects in these test.

Research paradigm

Positivism is the research paradigm most suited for this research study. This is due to the basic assumptions that the world is ordered and regular and it can be investigated objectively [6]. As we the researchers are a neutral part in the experiments, we are just trying to observe the subjects and we gain quantitative data from the experiment, we fall under the characteristics of positivism.

References

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NSD

The research method we chose for the research question involves getting user data from the participants, however the user data is limited to the time used finishing a set task. The data does need any personal information, so the participants are anonymous. Thus we concluded, as there's no need for personal data, that the NSD schema won't be necessary for the research. The most important part of the information is the time they use on tasks and not the people themselves.